

Impact of Generative AI on Software Engineering

ResearchAI RAG & Knowledge Graph Test Document

Overview

Generative AI is changing software engineering by assisting developers with code generation, debugging, testing, documentation, and technical research. AI coding assistants can reduce repetitive work and help developers explore solutions faster, while human review remains important.

Key Areas

- **Developer Productivity:** AI tools can accelerate code completion, refactoring, and documentation.
- **Code Quality:** AI-generated code can be useful but should be reviewed, tested, and validated.
- **Security:** AI-generated code may contain vulnerabilities or unsafe implementation patterns.
- **Research:** AI can summarize large amounts of information and organize research questions.
- **Future Impact:** AI agents may automate more parts of the software development lifecycle.

Important Risks

Major concerns include hallucinated information, insecure code, intellectual-property issues, privacy risks, over-reliance on AI, and difficulty verifying generated claims. Production systems should combine AI assistance with testing, source verification, access controls, and human oversight.

Research Questions

- 1 How does Generative AI affect developer productivity?
- 2 What security risks are introduced by AI-generated code?
- 3 How can RAG improve the reliability of AI answers?
- 4 How can knowledge graphs improve relationships between research concepts?
- 5 What role will autonomous AI agents play in future software engineering?

Test Keywords

Generative AI, software engineering, developer productivity, code generation, security, RAG, retrieval augmented generation, embeddings, vector database, Pinecone, Neo4j, knowledge graph, AI agents, research automation.